Physics 5A Discussion (Week 8)

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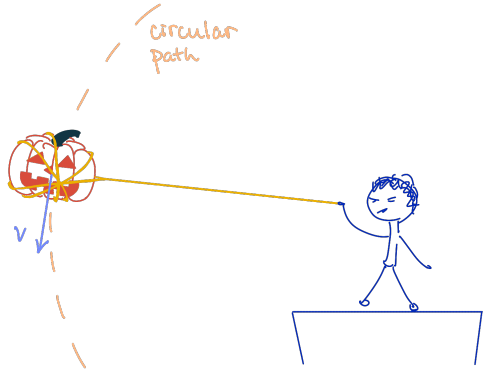
Jinyan is swinging a 2 kg pumpkin in a huge circular path by using a rope attached to the pumpkin. The radius of the circular path is $R = 1\,\text{m}$. He is trying to stop the pumpkin so he is making it spin slower. At time $t = 0$ seconds, the pumpkin is spinning with an angular velocity $\omega = 5$ rad/s counterclockwise. He is able to maintain a tangential acceleration $a_t = -1$ m/s^2.



Part (a) Find the centripetal acceleration and centripetal force that the pumpkin experiences at time $t = 0$ seconds. (*Hint: First find the tangential speed at time* $t = 0$.)

Part (b) Find the angular acceleration of the pumpkin.

Part (c) Find the tangential speed and angular speed of the pumpkin as a function of time (that is, leaving t as a variable in your final expression).

Part (d) Find the centripetal acceleration of the pumpkin as a function of time.

Part (e) Last week, we found that the pumpkin has to go faster than $v_t = 3.13$ m/s. Find the time that the pumpkin will reach $v_t = 3.13$ m/s.

Part (f) Find the frequency and period of the circular motion at time $t = 0$ seconds.

At time $t = 0$ seconds, the center of mass of the pumpkin is located at $\vec{r}_p = (1, 2)$ in meters, and the center of mass of Jinyan is located at $\vec{r}_m = (0, 5)$ in meters. The pumpkin is $m_p = 2\,\text{kg}$ and Jinyan is $m_j = 60\,\text{kg}$. Find the center of mass of the Jinyan-holding-the-pumpkin system, ignore the mass of the rope.

GOOD LUCK ON THE EXAM.

Solutions:

Part (a): The tangential speed at time $t = 0$ is $v_{t,0} = \omega R = 5\,\mathrm{m/s}$. Then we can find the centripetal acceleration at time $t = 0$ seconds,

$$a_c = \frac{v_t^2}{R} = 25\,\mathrm{m/s^2}\,.$$

Centripetal force at time $t = 0$ is obtained from Newton's second law,

$$F_c = ma_c = (2\,\mathrm{kg}) \cdot (25\,\mathrm{m/s^2}) = 50\,\mathrm{N}\,.$$

Part (b): The angular acceleration of the pumpkin is a constant,

$$\alpha = \frac{a_t}{R} = -1\,\mathrm{rad/s^2}\,.$$

Part (c): The tangential speed as a function of time is

$$v_t = v_{t,0} + a_t t = 5 - t\,,$$

and the unit is meter per second. The angular speed of the pumpkin as a function of time is

$$\omega = \omega_0 + \alpha t = 5 - t\,,$$

and the unit is rad per second. It is a coincidence here $v_t = \omega$ numerically (because radius is 1 meter), generally they have different numerical values.

Part (d): The centripetal acceleration of the pumpkin is not a constant, because its tangential speed is changing,

$$a_c = \frac{v_t^2}{R} = \frac{(5-t)^2}{1} = (5-t)^2\,,$$

with unit being $\mathrm{m/s^2}$.

Part (e) We have found the tangential speed v_t as a function of time, so we set $v_t = 3.13\,\mathrm{m/s}$ and find the time t,

$$v_t = 5 - t = 3.13\,,$$

so we get $t = 1.87$ seconds.

Part (f) This is the same problem that we did last week. The period is defined by the time it takes to go one full circle, which corresponds to the angle $\theta = 2\pi$, or the distance traveled $d = 2\pi R$ being the circumference of the circle. We can find the period T using either way, starting with the easy one, time is distance over velocity,

$$T = \frac{2\pi R}{v_t} = \frac{2\pi}{5}\,\mathrm{s} = 1.26\,\mathrm{s}\,.$$

Or using the angle, again, time is angle over angular velocity,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{5}\,\mathrm{s} = 1.26\,\mathrm{s}\,.$$

Frequency is computed by taking the inverse of period,

$$f = \frac{1}{T} = 0.793\,\mathrm{rev/s}$$

Center of mass is calculated using the following formula

$$\vec{r}_{\mathrm{COM}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} = \frac{m_p \vec{r}_p + m_j \vec{r}_j}{m_p + m_j} \,.$$

This involves adding vectors, so we need to break it down to components.

$$\vec{r}_{\mathrm{COM}} = \left(\frac{(2\,\mathrm{kg}) \cdot (1\,\mathrm{m}) + (60\,\mathrm{kg}) \cdot (0\,\mathrm{m})}{(2\,\mathrm{kg}) + (60\,\mathrm{kg})} \,,\ \frac{(2\,\mathrm{kg}) \cdot (2\,\mathrm{m}) + (60\,\mathrm{kg}) \cdot (5\,\mathrm{m})}{(2\,\mathrm{kg}) + (60\,\mathrm{kg})} \right) = \left(\frac{1}{31}\,\mathrm{m} \,,\ \frac{152}{31}\,\mathrm{m} \right) \,.$$